César Antonio Carvajal de la Haza

Mathematician · Statistician · Physicist

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QUALIFICATION SUMMARY

Mathematics, Statistics and Physics graduate with hands-on research experience in quantum many-body systems and entanglement entropy. Currently working as a Consulting Analyst specializing in process automation, and starting an M.Sc. in Quantum Technologies (QTEP) in October 2026. Combines a strong mathematical foundation with practical programming skills in Python, C/C++, and SQL, developed through research projects, algorithmic problem-solving, and applied machine learning. Particularly interested in quantum computing and high-performance scientific computing.

EDUCATION

M.Sc. in Quantum Technologies (QTEP) (Starting October 2026)

Universidad Internacional Menéndez Pelayo (UIMP), Spain

- Admitted to the QTEP Master's programme, ranked 2nd on the admissions list.
- Core focus on quantum computing, open quantum systems and quantum thermodynamics, quantum sensors, and quantum technologies laboratory work.

Bachelor's Degree in Physics — 240 ECTS (2021 – 2025) | GPA: 8.1/10

National University of Distance Education (UNED), Spain

- Strong background in classical and quantum physics, optics, and dynamical systems.
- Fields of interest: Quantum Physics, General Relativity, Theoretical Physics, Nuclear Physics, Optics.
- Honours in Classical Mechanics, Vibrations & Waves, Quantum Physics II, and Dynamical Systems.
- Final Grade Project: *Entanglement Entropy and Geometry in Quantum Systems* — see full description under Academic / Personal Projects.

Bachelor's Degree in Mathematics — 240 ECTS (2016 – 2021) | GPA: 8.0/10

University of Seville, Spain

- Strong foundation in calculus, algebra, and differential equations.
- Fields of interest: Computing, Algebra, Ordinary & Partial Differential Equations, Topology, Probability, Coding Theory & Cryptography.
- Gained experience in Haskell, Python, Maxima, and MATLAB; developed solid programming fundamentals through mathematical problem-solving.
- Particular interest in Algebra and the structures formed by number systems.

Bachelor's Degree in Statistics — 240 ECTS (2016 – 2021) | GPA: 8.3/10

University of Seville, Spain

- Expertise in probability, statistical inference, and data analysis.
- Fields of interest: Statistical Inference, Databases, Survey Design, Operations Research, Multivariate Analysis, Computational Statistics, Decision Theory, AI & Statistics.
- Developed multiple projects in R and Excel; introduced to Machine Learning algorithms throughout the programme.
- Honours in Survey Design; participated in a real-world user-satisfaction study at the University of Seville.

TRAINING & CONTINUING EDUCATION

Introduction to Quantum Computing — 1.5 ECTS (2026 – ongoing)

UNED, Spain · Extension course

- Foundational concepts in quantum computing: qubits, quantum gates, superposition, and entanglement. Covered core algorithms including Grover's and Shor's, introduction to Boltzmann Machines (BQM), and hands-on implementation using PennyLane.

GNSS Academy — Intensive GNSS Training Programme (January 2025 – July 2025)

GNSS Academy, Spain · 6-month combined study and practical programme

- Comprehensive study of Global Navigation Satellite System (GNSS) theory, architectures, and signals.

- Topics covered: DGNSS, RTK, PPP, SBAS, GNSS data processing, ionospheric and tropospheric corrections, multi-path mitigation, satellite orbit propagation (SP3), and advanced positioning techniques.
- Hands-on use of gLAB for analysing GNSS observables and signals.
- **Capstone project:** Full-featured GNSS+IMU sensor fusion simulator developed in Python/Linux:
  - Simulated GNSS measurements using precise SP3 orbit propagation and realistic error models (ionospheric, tropospheric, clock, multipath).
  - Applied Antenna Phase Offset (APO) and Sagnac effect corrections.
  - Implemented mechanisation equations in ECEF frame to derive position, velocity, and attitude from simulated IMU data.
  - Performed coordinate transformations for accurate motion tracking.
  - Implemented a Least-Squares Estimation module for GNSS-only positioning.

WORK EXPERIENCE

Consulting Analyst — RPA (June 2026 – present)

Minsait (Indra Group), Spain

- Joined the Data, Intelligence & Robots unit, specialising in Robotic Process Automation (RPA).
- Currently completing the UiPath Automation Developer Professional Training as part of onboarding.

Secondary School Teacher — Mathematics (January 2022 – June 2026)

Instituto de Educación Secundaria, Spain

- Teaching Mathematics from 3º ESO through 2º Bachillerato, covering algebra, calculus, statistics, and geometry.
- Preparation of exams, tests, and learning materials adapted to each educational level.
- Curriculum planning and organisation; design of personalised adaptations for students with diverse learning needs.
- Active participation in departmental meetings and academic committees.
- Providing academic guidance and study-skills coaching to students.

QA Automation Engineer (April 2022 – November 2022)

Deloitte, Spain

- Developed automated test suites using Selenium and Java, significantly reducing manual testing time.
- Co-implemented an automation framework using Selenium, Cucumber, and Jenkins, improving coverage and efficiency.
- Executed test scripts, documented results, and created detailed bug reports.
- Identified and prioritised defects using Jira; maintained synchronised codebase with Git.
- Collaborated with cross-functional teams in an international environment.

Private Tutor — Mathematics, Physics, Chemistry & Technical Drawing (several years)

Self-employed

- Tutored students across multiple subjects, improving their academic performance and confidence.
- Created a supportive environment focused on clarity, logical reasoning, and problem-solving.

RESEARCH & ACADEMIC / PERSONAL PROJECTS

MLWeather — Temperature Forecasting with ML (2026)

Personal project · GitHub · Live app

- End-to-end pipeline for temperature forecasting in Seville using hourly meteorological data from Open-Meteo API.
- Compared 5 models: Linear Regression, XGBoost, Random Forest, SARIMA, and LSTM; evaluated with MAE, RMSE, and R^2.
- Applied preprocessing techniques including lag features, cyclical time encodings (sin/cos), and strict train/test separation to avoid data leakage.
- Built an interactive Streamlit app with modular architecture (src/, models/, ui/, evaluation/) for exploring and comparing model predictions.

Algorithmic Problem Solving — LeetCode (2026 – present)

Self-directed · GitHub

- Solving algorithmic and database problems across all difficulty levels (Easy to Hard).
- Practising data structures and algorithms in C++: arrays, linked lists, two-pointer techniques, binary search, and dynamic programming.

- Solving SQL problems covering joins, aggregations, and query optimisation.
- Solutions tracked and version-controlled on GitHub.

Terminal & Shell Scripting (ongoing)

Self-directed

- Practising command-line workflows in a Unix/Linux environment: file and directory manipulation (mkdir, touch, mv, cp), text search and pattern matching (grep), and basic Bash scripting.
- Writing and executing .sh scripts to automate command-line tasks.

Entanglement Entropy and Geometry in Quantum Systems (2024 – 2025)

Final degree project — Physics · GitHub | Outreach article (UNED Physics Blog)

- Investigated the relationship between physical interaction geometry and entanglement structure in systems of hard-core bosons, testing the validity of the entanglement Area Law.
- Modelled hopping interactions using a Hubbard Hamiltonian on graphs with varying topologies (periodic chains, bridge, and tree geometries), constructing the system's adjacency matrix from physical connectivity.
- Computed ground states via numerical diagonalisation and derived reduced density matrices to obtain pairwise entanglement entropies.
- Built the Entanglement Adjacency Matrix (EAM), comparing physical connectivity against quantum correlation structure to identify Area Law violations and entanglement monogamy effects.
- Implemented all simulations and numerical analysis in Python.
- Note: findings were later adapted into a public outreach article for the UNED Physics Department blog.

Science Outreach — “Entre grafos y entrelazamiento” (2025)

Personal project · UNED Physics Blog

- Wrote a public-facing article explaining the undergraduate thesis research on entanglement entropy for a general academic audience, as part of the UNED Physics Department's outreach blog.

Survey Design & Statistical Analysis (2019/20)

University of Seville — Statistics degree

- Participated in the 2019/20 User Satisfaction Study of the IT Service at the University of Seville.
- Contributed to survey design, data collection, statistical analysis in R and Excel, and final report preparation.

Problem-Solving & Coding Challenges (2016 – 2018)

Self-directed

- Solved algorithmic problems on Exercitium and Kattis to develop problem-solving and computational thinking skills.

TECHNICAL SKILLS

Programming Languages: Python, R, Haskell, MATLAB, SQL (intermediate), C/C++ (intermediate), Bash (basic)

Tools & Platforms: Linux, Git, VSCode, Jupyter, LaTeX, gLAB

GNSS & Navigation: DGNSS, RTK, PPP, SBAS, GNSS data processing, SP3 orbit propagation, IMU sensor fusion, ECEF transformations

Data Analysis & Modelling: Applied statistics, mathematical modelling, machine learning, optimisation

Applied Mathematics: Algebra, differential equations, cryptography, topology, optimisation

Quantum: Quantum mechanics, entanglement entropy, quantum computing (introductory)

SOFT SKILLS

Critical Thinking · Analytical Skills · Creative Thinking · Attention to Detail · Adaptability · Research Skills · Interpersonal Skills

LANGUAGES

Spanish (Native) · English (C1 — Proficient) · French (Basic, in progress)

EXTRACURRICULAR ACTIVITIES & VOLUNTEERING

LGTB+ Advocacy & Awareness (2017 – 2019)

Colectivo LGTB+ Pedro Zerolo, Faculty of Mathematics, University of Seville

- Participated in awareness campaigns and outreach activities related to the LGBT+ community.

Animal Welfare Volunteering

Local animal shelter

- Provided weekly care for rescued dogs; created adoption advertisements and participated in awareness campaigns.

INTERESTS

Quantum Physics · Artificial Intelligence · Data Visualisation · Language Learning · Writing · Crossfit